Ethan Wayne Holbrook

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PROFESSIONAL SUMMARY

Materials Engineering Ph.D. with expertise in atomistic modeling, HPC workflow development, FAIR workflow creation and automation. Research work focused on understanding multi-component energetic materials. Knowledge base in materials science & engineering.

EDUCATION

Doctor of Philosophy – Materials Engineering *2021 – 2026*
Computational Engineering Concentration || *Purdue University*, West Lafayette, IN

Bachelor of Science – Materials Science & Engineering *2017 – 2021*
Mathematics Minor || *Wright State University*, Dayton, OH

PROFESSIONAL EXPERIENCE

Graduate Research Assistant *2021 – Present*
Purdue University, West Lafayette, IN

- Led atomistic simulations of shock-driven jetting in multi-component high explosives, showing how shock strength, crystallography, and interfacial structure govern jet formation, mixing, and localized heating relevant to hotspot formation.
- Built a parsing and validation framework for simulation inputs, including systematic evaluation of LLM-generated scripts to identify failure modes and support more autonomous molecular dynamics workflows.
- Investigated high-strain-rate deformation in aluminum single crystals using molecular dynamics and OVITO, characterizing defect evolution, plasticity, and microstructural response under extreme loading.

LLNL Intern (CCMS/MaCI) May-Aug. *2022 – 2024*
Lawrence Livermore National Laboratory, Livermore, CA

- Performed force-field validation using equation-of-state and vibrational spectra / normal mode analyses, helping establish fidelity across thermodynamic and molecular-response metrics.
- Combined validated TNT and HMX force fields into a unified framework for atomistic simulation of Octol, enabling molecular-level investigation of multi-component energetic materials.
- Developed computational workflows in LAMMPS for constructing arbitrary molecular-crystal interfaces.

AFRL RXAP SOCHE Intern *2020 – 2021*
Air Force Research Laboratory, Dayton, OH

- Supported setup of a laser-heated pedestal growth lab by organizing equipment and preparing spaces for research use and developing custom lab components in SolidWorks.
- Measured thin-film optical properties using spectroscopic ellipsometry and refined YAG fiber polishing procedures to meet analysis requirements.

RELEVANT SKILLS

Programming Languages

- Proficient: Python, LaTeX
- Familiar: Bash, Fortran90, MATLAB
- Tools: HPC, SLURM, Git

Machine Learning and Data Science

- Frameworks: TensorFlow/Keras, PyTorch
- Libraries: Pandas, Scikit-learn, Plotly

Software

- Computational Packages: LAMMPS, Quantum ESPRESSO
- Analysis: OVITO, Avogadro
- Design: SolidWorks
- Others: LabVIEW, Microsoft Office Suite